

SQL QUERY

```
use project_orders;
```

```
select
    a.aisle,
    count(p.product_id) as total_products
from products p
join aisles a
on p.aisle_id = a.aisle_id
group by a.aisle
order by total_products desc
limit 10;
```

```
select
    count(distinct department_id) as total_departments
from departments;
```

```
select
    d.department,
    count(p.product_id) as total_products
from products p
join departments d
on p.department_id = d.department_id
group by d.department
order by total_products desc;
```

```
select
```

```
p.product_name,  
round(avg(opt.reordered) * 100, 2) as reorder_rate  
from order_products_train opt  
join products p  
on opt.product_id = p.product_id  
group by p.product_id, p.product_name  
order by reorder_rate desc  
limit 10;
```

```
select  
count(distinct user_id) as total_users  
from orders;
```

```
select  
user_id,  
round(avg(days_since_prior_order), 2) as average_days_between_orders  
from orders  
where days_since_prior_order is not null  
group by user_id  
order by average_days_between_orders;
```

```
select  
order_hour_of_day,  
count(order_id) as total_orders  
from orders  
group by order_hour_of_day  
order by total_orders desc;
```

```
select
    order_dow,
    count(order_id) as total_orders
from orders
group by order_dow
order by order_dow;
```

```
select
    p.product_name,
    count(opt.product_id) as total_orders
from order_products_train opt
join products p
on opt.product_id = p.product_id
group by p.product_id, p.product_name
order by total_orders desc
limit 10;
```

```
select
    d.department,
    count(distinct o.user_id) as total_users
from orders o
join order_products_train opt
on o.order_id = opt.order_id
join products p
on opt.product_id = p.product_id
join departments d
on p.department_id = d.department_id
group by d.department
```

```
order by total_users desc;
```

```
select
```

```
    round(avg(product_count), 2) as average_products_per_order
```

```
from
```

```
(
```

```
    select
```

```
        order_id,
```

```
        count(product_id) as product_count
```

```
    from order_products_train
```

```
    group by order_id
```

```
) as order_summary;
```

```
select
```

```
    d.department,
```

```
    p.product_name,
```

```
    sum(opt.reordered) as reorder_count
```

```
from order_products_train opt
```

```
join products p
```

```
on opt.product_id = p.product_id
```

```
join departments d
```

```
on p.department_id = d.department_id
```

```
group by d.department, p.product_name
```

```
order by d.department, reorder_count desc;
```

```
select
```

```
    count(*) as products_reordered_more_than_once
```

```
from
```

```
(  
    select  
        product_id  
    from order_products_train  
    where reordered = 1  
    group by product_id  
    having count(*) > 1  
) as reordered_products;
```

```
select  
    round(avg(max_cart_position), 2) as average_products_added_to_cart  
from
```

```
(  
    select  
        order_id,  
        max(add_to_cart_order) as max_cart_position  
    from order_products_train  
    group by order_id  
) as cart_summary;
```

```
select  
    order_hour_of_day,  
    count(order_id) as total_orders  
from orders  
group by order_hour_of_day  
order by order_hour_of_day;
```

```
select
```

```
    product_count,  
    count(*) as total_orders  
from  
(  
    select  
        order_id,  
        count(product_id) as product_count  
    from order_products_train  
    group by order_id  
) as order_summary  
group by product_count  
order by product_count;
```

```
select  
    a.aisle,  
    round(avg(opt.reordered) * 100, 2) as average_reorder_rate  
from order_products_train opt  
join products p  
on opt.product_id = p.product_id  
join aisles a  
on p.aisle_id = a.aisle_id  
group by a.aisle  
order by average_reorder_rate desc;
```

```
select  
    o.order_dow,  
    round(avg(order_size), 2) as average_order_size  
from
```

```
(  
    select  
        order_id,  
        count(product_id) as order_size  
    from order_products_train  
    group by order_id  
) as order_summary  
join orders o  
on order_summary.order_id = o.order_id  
group by o.order_dow  
order by o.order_dow;
```

```
select  
    user_id,  
    count(order_id) as total_orders  
from orders  
group by user_id  
order by total_orders desc  
limit 10;
```

```
select  
    d.department,  
    a.aisle,  
    count(p.product_id) as total_products  
from products p  
join aisles a  
on p.aisle_id = a.aisle_id  
join departments d
```

```
on p.department_id = d.department_id  
group by d.department, a.aisle  
order by d.department, total_products desc;
```